

- 1 (a) Simplify $\frac{2x}{x^0} \div \left(\frac{2y}{x}\right)^{-2}$, leaving your answer in positive indices.

Answer _____ [3]

- (b) Express y in terms of p and q , given that $\frac{1}{q} = \frac{2}{y} + p$.

Answer _____ [3]

- 2 (a) Express $\frac{1}{x-3} - \frac{6}{x^2-9}$ as a single fraction in its simplest form.

Answer _____ [2]

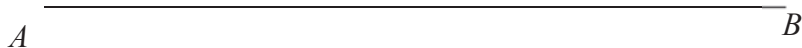
- (b) (i) Express $y = x^2 + 4x - 5$ in the form $y = (x + a)^2 + b$, where a and b are constants.

Answer _____ [2]

- (ii) Write down the equation of the line of symmetry of the graph of $y = x^2 + 4x - 5$.

Answer _____ [1]

- 3 The line AB is shown below.
- (a) Construct an equilateral triangle ABC . [1]
 - (b) Construct the perpendicular bisector of AB . [1]
 - (c) Construct the angle bisector of $\angle CAB$. [1]
 - (d) Mark clearly a possible point that is inside the triangle, equidistant from AC and AB , and is nearer to A than B . Label this point P . [1]



- (e) The point Q is such that $\angle ACQ = 120^\circ$ and $\frac{BQ}{BA} = 0.9$.
 Find two possible positions of Q . Label these point Q_1 and Q_2 .
 Measure and write down the length AQ (where Q can be any of the two points).

Answer _____ cm [3]